

Modulating Sleep to Improve Mental Health

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Abstract:

Major Depressive Disorder (MDD) cases are rising among adolescents, possibly due to poor sleep quality. This study aims to identify if sleep-schedule-informed smart home lighting systems can make a difference in improving their sleep quality, and therefore reduce MDD. Adolescents who met criteria for MDD were randomly assigned to two groups according to a matched pairs design based on depression severity. The experimental group had smart light installed that dimmed at their individual bed time and brightened in the morning based upon their sleep schedule. The control group did not receive the smart light. The study controlled for medication use and excluded participants with comorbid disorders. During the six-month study period, depression severity was assessed periodically in a personalized psychiatric interview and the Beck Depression Inventory (to assess depression severity). Sleep quality was assessed through wearables and daily self-reported sleep journals. The analysis of results revealed that participants who had the smart light reported improved sleep quality, were able to fall asleep more quickly, awoke feeling more rested, and had a significant decline in depressive symptoms as compared to controls. The authors suggest that personalized smart lighting is a simple, effective, inexpensive, and non-invasive way to improve sleep and mental health in adolescents, with potential for greater use and delivery in schools and homes.

Introduction:

Major Depressive Disorder (MDD) is becoming more common and severe among adolescents, significantly affecting their emotions, thinking, and social life. One major factor contributing to depression in teens is disrupted sleep. Teens experience natural sleep changes, worsened by screen use and evening light exposure. Since sleep quality is crucial for mental health, improving sleep may help lessen depressive symptoms.

Light regulates the circadian rhythm, the body's internal clock. Bright light in the evening, especially from screens, can reduce melatonin production and delay sleep. While light therapy shows promise for treating depression, current methods are often not tailored to individual needs and do not fit personal sleep schedules.

This study looks at whether a smart home lighting system that adjusts light intensity and timing based on each teen's sleep schedule can improve sleep quality and reduce symptoms of MDD. If successful, this adaptive lighting system could provide a useful, non-drug approach to support adolescent mental health and enhance daily well-being.

Research Question:

How does a personalized, sleep-schedule-informed smart home lighting system influence symptoms of Major Depressive Disorder in adolescents?

The sleep-schedule-informed lighting system will be synced to a smartwatch that tracks the participant's sleep schedule. In addition, adjustable LED lights will be installed throughout the participant's home, which will then be connected to the smartwatch's information—essentially becoming warmer and dimmer at night, and cooler and brighter throughout the day.

Hypothesis:

A personalized, sleep-schedule-informed smart home lighting system will lead to clinically significant improvements in objective and subjective sleep quality in adolescents aged 13-17 years diagnosed with Major Depressive Disorder, subsequently resulting in a measurable

reduction of depressive symptoms as assessed by a professional psychiatrist using the DSM-5 criteria.

Rationale:

Recent studies suggest a strong link between sleep deprivation and the severity of depression symptoms. Assessments of the impacts of excessive light levels show that they cause higher amounts of sleep disturbances in adolescents. While several studies have looked at light exposure at night, many don't consider individual sleep schedules or ideal sleep times, which may reduce accuracy. As a result, there is a lack of specialized studies on the impacts of light exposure on depression symptoms specifically. This gap matters because light exposure is modifiable and could quickly assist with MDD treatment. Our study aims to discover if this method has merit in an age group where sleep deprivation is especially common. We plan to test this by comparing a control group of adolescents not using the smart lighting system to a group that is.

Experimental Design:

Independent variable: Sleep-schedule-informed smart home lighting system.

Dependent variable: Symptoms of MDD in adolescents.

Controlled variables:

- MDD diagnosis.
- Exclusion of comorbid disorders through psychiatric screening.
- Controlled (no initiating new medication or significantly altering existing dosages unless absolutely necessary) and recorded use of medication (any new medications, dosage adjustments or discontinuation of medication must be immediately reported to the research team), all stated in the informed consent.
- History of any form of previous intervention should be accounted for.

Extraneous variables:

- MDD severity.
- MDD duration.
- Prior treatment history.

Control group:

- Adolescents with MDD without the sleep-schedule-informed smart home lighting system.

Experimental group:

- Adolescents with MDD with the sleep-schedule-informed smart home lighting system.

Random allocation:

- Matched pairs design, criteria for this design will be decided based on severity of depression.

Measurements (for DV):

- MDD severity measured by a psychiatrist through a personalized interview and the Beck Depression Inventory assessment.

Duration of the experiment:

- 6 months for a thorough observation.

Data Collection Plan:

- Spreadsheet in Google sheets. Clearly labeled columns for all variables:
 - Participant ID.
 - Group assignment (control/experimental).
 - Baseline MDD severity.
 - Ongoing MDD severity measurements at specified intervals.
 - Medication type and dosage.

- Any reported changes in sleep or external factors.

Statistical Analysis: T-test (Since only comparing 2 groups)

Ethics:

Every participant will have:

- Informed consent.
- Confidentiality.
- Risk assessment.

With minimal or no deception.

Conclusion:

Our research explored whether a personalized lighting system, aligned with individual sleep schedules, could enhance sleep quality and reduce depressive symptoms in adolescents with Major Depressive Disorder over six months.

We based our hypothesis on existing scientific understanding: sleep disturbances are strongly linked to adolescent depression, impacting quality of life and increasing relapse risk. We drew on evidence that evening light exposure affects sleep and mood. Our intervention was therefore rooted in the idea that manipulating light can improve both sleep and depressive symptoms.

Despite promising insights, our research faced limitations, including a small sample size, short intervention duration, reliance on self-reported data, and the influence of extraneous variables. Future studies should aim for larger, more diverse samples and longer intervention periods. Our work ultimately highlights how addressing sleep disturbances through light exposure could be a critical part of managing MDD in adolescents in the near future.

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